

SANGJUN (SAMUEL) Ko

202 E White St, Champaign, Illinois, 61820 | sangjun2@illinois.edu | www.linkedin.com/in/sangjunko

EDUCATION

University of Illinois at Urbana-Champaign Bachelor of Science in Statistics (CGPA: 3.83/4.00) Mathematics & Data Science Minor	Champaign, Illinois January 2023 - May 2025 (Expected)
University of Michigan-Shanghai Jiao Tong University Joint Institute Bachelor of Engineering in Mechanical Engineering (CGPA: 3.50/4.00) First Class Scholarship	Shanghai, China September 2019 - December 2020

WORK EXPERIENCE

STAT107 Data Science Discovery <i>Undergraduate Course Assistant</i> - Facilitate learning for over 100 students by assisting students in grasping fundamental data science concepts by conducting interactive sessions, and providing clarifications on course materials - Manage Git for learning platform modifications, benefiting 1000+ students; create, edit, and proofread course materials - Collaborate with a team of 100+ staff and 2 professors in weekly meetings	Champaign, Illinois August 2023 - Present
STAT443 Professional Statistics <i>Undergraduate Course Assistant</i> - Grade assignments and provide feedback to help 20+ students improve statistical analysis and consulting skills - Evaluate 5+ student presentations to ensure alignment with course expectations and grading criteria - Act as a client in mock consulting projects, providing insights and posing real-world challenges	Champaign, Illinois January 2025 - Present
Applied Technologies for Learning in Arts & Sciences <i>AI/ML Intern Team Leader</i> - Lead a team of 5 interns - Developed predictive machine learning models with large data sets(100,000+ entries) using Python - Utilized advanced techniques: XGBoost, Random Forests, Regression, Neural Networks, BERT, Transformers - Collaborated closely with 3 coordinators in weekly meetings to align project goals and track progress	Champaign, Illinois January 2024 - December 2024
KM Internacional S.A. <i>Intern</i> - Oversaw comprehensive inventory management and streamlined administrative tasks for a small textile company - Lead a team of three workers to achieve a 98% accuracy rate in record-keeping and a 15% reduction in operational costs	Managua, Nicaragua May 2023 - August 2023

PROJECTS/ AWARDS

Graduation Prediction Model - Increased AUC score from 0.65 to 0.82 by reducing noise in the dataset through feature engineering and outlier removal across 50,000+ student records. - Built an interactive Streamlit interface with 12 key predictors, enabling real-time graduation predictions for university administrators.	September 2024 - December 2024
3rd Place, 2024 COVID Information Commons Student Paper Challenge - Analyzed the socioeconomic and mental health impacts of COVID-19 in Mexico, identifying that 42% of females and 40% of young adults faced job loss - Proposed data-driven interventions to support vulnerable groups through geospatial and statistical analysis	September 2024
Paraphrase Detection - Achieved 90%+ accuracy in paraphrase detection using BERT, RoBERTa, and SBERT models on the MRPC dataset - Developed a Gradio interface for real-time paraphrase detection, facilitating easier evaluation of textual similarities for academic and business use cases	July 2024

SKILLS

Languages:	English (Fluent), Spanish (Fluent), Korean (Fluent), French (Basic), Chinese (Basic)
Computer:	Python (Advanced): Pandas, NumPy, Scikit-learn, TensorFlow, NLTK, Matplotlib, Seaborn, BERT, Random Forest, XGBoost, Gradio R (Intermediate): Tidyverse, ggplot2, dplyr, caret Git (Intermediate), SAS (Basic), Tableau (Basic), SQL (Basic), Office (Advanced)
Certificates:	Data Science DISCOVERY, Data Science Certificate, Coursera Business Analytics Certificate

COMMUNITY SERVICE AND EXTRACURRICULAR ACTIVITIES

Consultant in Illinois Business Consulting

September 2024 - December 2024

- Collaborating with interdisciplinary teams to deliver strategic solutions for real-world business challenges, including market analysis, operational optimization, and growth strategy, for clients ranging from Fortune 100 companies to nonprofits